BF28.6

1. The flux through the inductor to the right is decreasing as the magnet moves, so the induced magnetic field should point towards the magnet. This means the current moves counterclockwise through the diagram. When the magnet is at rest, there no is no change in flux and no current through the wire. When the magnet moves left out of the wire, there is decreasing leftwards flux through the coil, inducing a right to left current through the meter.
2. Knowns: $l = 0.1\ \mathrm{m}, R = 0.35\ \Omega, v = 4\ \mathrm{m/s}, P = 3.6\ \mathrm{W}$
   Unknowns: $F = ?, B = ?$

$$\begin{aligned} V &= \sqrt{RP} \\ &= \sqrt{(0.35\ \Omega)(3.6\ \mathrm{W})} \\ &= 1.122\ \mathrm{V} \\ V &= Blv \\ B &= \frac{V}{lv} \\ &= \frac{1.122\ \mathrm{V}}{(0.1\ \mathrm{m})(4\ \mathrm{m/s})} \\ &= \boxed{2.805\ \mathrm{T}} \\ F = F_b &= IlB \\ &= \left(\frac{1.122\ \mathrm{V}}{0.35\ \Omega}\right)(0.1\ \mathrm{m})(2.81\ \mathrm{T}) \\ &= \boxed{0.9\ \mathrm{N}} \end{aligned}$$

3. Knowns: diagram
   Unknowns: $\Phi_m = ?$

$$\begin{aligned} \Phi_m &= BA\cos\theta \\ &= (7 \cdot 10^{-2}\ \mathrm{T})(0.1\ \mathrm{m})^2 \cos(45^\circ) \\ &= \boxed{0.000\,495\ \mathrm{Wb}} \end{aligned}$$

4. Knowns: diagram
   When Magnet 1 moves right, the induced field should point left. This means the current should move right to left across R. Since the second magnet is perpendicular to the inductor, there is no induced current.
5. Knowns: diagram
   Unknowns: $\Phi_m = ?, direction$

$$
\begin{aligned}
\Phi_m &= BA \\
&= \frac{1}{2}(0.25\ \mathrm{T})(0.1\ \mathrm{m})(0.1\ \mathrm{m} * \sqrt{3}) \\
&= \boxed{0.00217\ \mathrm{Wb}}
\end{aligned}
$$

By Lenz's Law, if the magnetic field strength decreases, the current flows clockwise.

6. Knowns: $Figure_{1,2,3}, r = 0.075\ \mathrm{m}, R = 0.22\ \Omega$

   Unknowns: $I_{1,2,3} = ?, direction_{1,2,3} = ?$

$$
\begin{aligned}
\varepsilon &= -\frac{d\Phi_m}{dt} \\
I &= -\frac{A}{R}\frac{dB}{dt} \\
|I_1| &= \frac{\pi(0.075\ \mathrm{m})^2}{0.22\ \Omega}(0.5\ \mathrm{T/s}) \\
&= 0.0402\ \mathrm{A} \\
direction &= cw \\
|I_2| &= \frac{\pi(0.075\ \mathrm{m})^2}{0.22\ \Omega}(0.5\ \mathrm{T/s}) \\
&= 0.0402\ \mathrm{A} \\
direction &= cw
\end{aligned}
$$

   $|I_3| = 0\ \mathrm{A}$ and there is no induced current because the magnetic field is perpendicular to the normal vectors of the surface.

7. Knowns: $R = 0.1\ \Omega$, diagram

   Unknowns: $\mathrm{sign}\left(\frac{dB}{dt}\right) = ?, \left|\frac{dB}{dt}\right| = ?$

   Since the induced current produces a field pointed out of the page, the net flux is increasing into the page.

$$
\begin{aligned}
\left|\frac{dB}{dt}\right| &= \frac{IR}{A} \\
&= \frac{(0.15\ \mathrm{A})(0.1\ \Omega)}{(0.08\ \mathrm{m})^2} \\
&= \boxed{2.34\ \mathrm{T/s}}
\end{aligned}
$$

8. Knowns: $N = 100, r = 0.045\ \mathrm{m}, r = 0.00025\ \mathrm{m}, I = 4\ \mathrm{A}$

   Unknowns: $\left|\frac{dB}{dt}\right| = ?$

$$
\begin{aligned}
\varepsilon &= NA\left|\frac{dB}{dt}\right| \\
R &= \rho\frac{l}{A} \\
&= (1.7\cdot 10^{-8}\ \Omega{\cdot}\mathrm{m})\frac{100(2\pi(0.045\ \mathrm{m}))}{\pi(0.000\,25\ \mathrm{m})^2} \\
&= 2.448\ \Omega \\
\left|\frac{dB}{dt}\right| &= \frac{IR}{NA} \\
&= \frac{(4\ \mathrm{A})(2.448\ \Omega)}{(100)(\pi(0.045\ \mathrm{m})^2)} \\
&= \boxed{15.4\ \mathrm{T/s}}
\end{aligned}
$$

9. Knowns: $A = 0.0256\ \mathrm{m}^2, B = (0.33t\hat{\imath} + 0.48t^2\hat{k})T$
   Unknowns: $\varepsilon(0.5) = ?, \varepsilon(1) = ?,$

$$
\begin{aligned}
\varepsilon(t) &= \frac{d\Phi_m}{dt} \\
&= \frac{dB}{dt}A\sin(\theta) \\
&= (0.96t)(0.0256\ \mathrm{m}^2) \\
\varepsilon(0.5) &= \boxed{0.0123\ \mathrm{V}} \\
\varepsilon(1) &= \boxed{0.0246\ \mathrm{V}}
\end{aligned}
$$

10. Knowns: $s = 0.2\ \mathrm{m}, R = 0.47\ \Omega, B = 0.81\ \mathrm{y}^2\ \mathrm{t}, t = 0.54\ \mathrm{s}$
    Unknowns: $I = ?$

$$
\begin{aligned}
\Phi_m &= \int B\cdot dA \\
&= \int_0^{0.2}\int_0^{0.2}(0.81y^2t)dydx \\
&= (0.2)\left[0.27y^3t\right]_{y=0}^{y=0.2} \\
&= 0.000432t \\
\varepsilon &= -\frac{d\Phi_m}{dt} \\
&= 0.000432 \\
I &= \frac{V}{R} \\
&= \frac{0.000\,432\ \mathrm{V}}{0.47\ \Omega} \\
&= \boxed{0.000\,919\ \mathrm{A}}
\end{aligned}
$$